

Structural Admissibility: An I2OS Framework for Secure Generative Systems

Anonymous · 2026

Abstract

We introduce Structural Admissibility, a security framework for generative systems based on transition validity rather than post-generation filtering. Unlike conventional AI safety mechanisms that filter outputs after generation, the proposed I2OS framework permits execution only when a state transition is structurally admissible. The framework integrates admissibility constraints, reality confirmation, and collapse detection. We further present a use case for LLM hallucination suppression by reformulating hallucination as an unsupported state transition.

1. Introduction

Modern AI systems rely on optimization, prediction, and post-hoc filtering for safety. Structural Admissibility shifts safety from output-level filtering to transition-level constraints.

2. Structural Admissibility

A transition is admissible when $C(S_t, T, S_{t+1}) = 1$. Execution occurs only when $GO = 1[C(S_t, T, S_{t+1}) = 1]$.

3. I2OS Security Model

$GO(t) = 1[\Gamma(Q(t))R(t) > \theta \text{ AND } Reality(t) = 1 \text{ AND NOT } Collapse(t)]$. Security is defined as non-collapse under admissible transition.

4. Architecture

Input → State Mapping → Transition Generation → Admissibility Check → Reality Confirmation → Execution → Collapse Detection → Alignment Logging.

Structural Admissibility Architecture

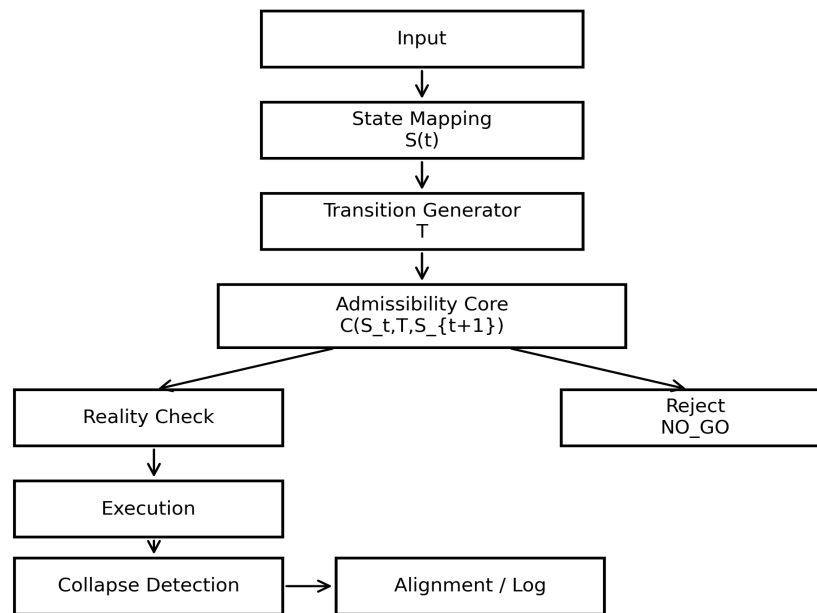


Figure 1. Structural Admissibility Architecture.

Conventional vs I2OS Safety

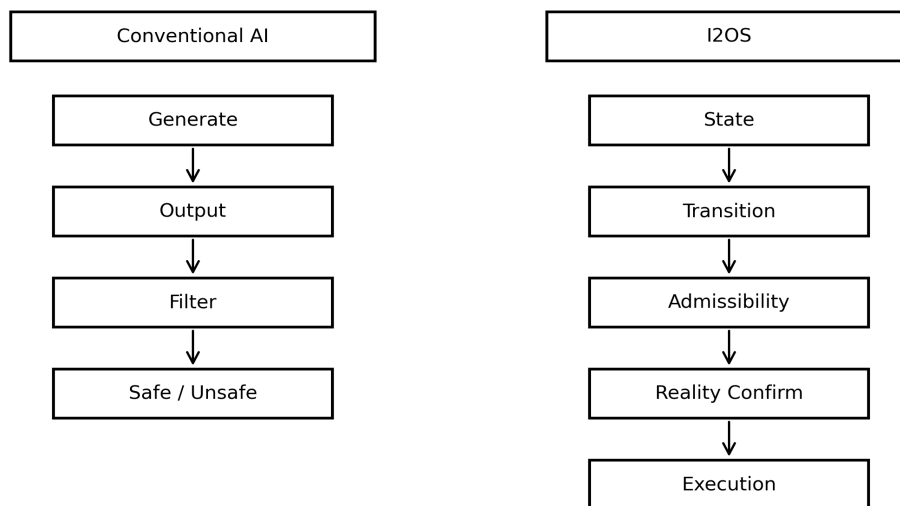


Figure 2. Conventional vs I2OS Safety.

Admissibility Funnel

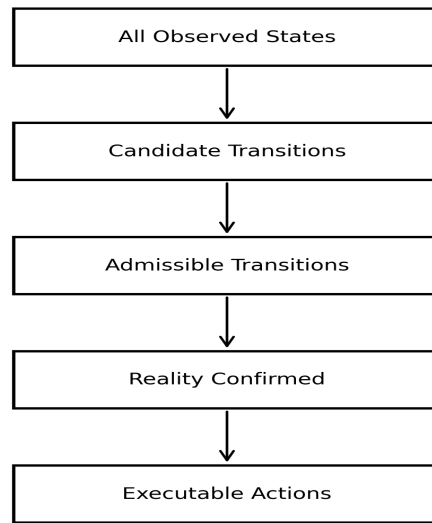


Figure 3. Admissibility Funnel.

Admissible vs Collapse Region

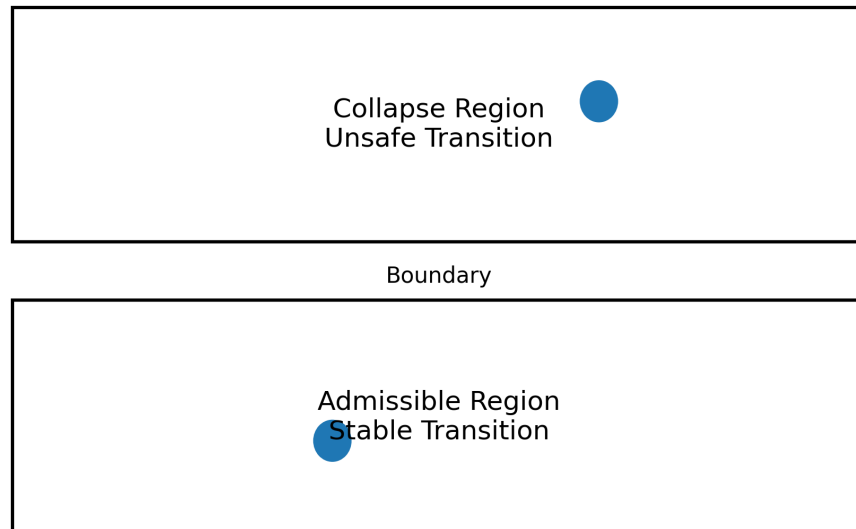


Figure 4. Admissible vs Collapse Region.

5. Experimental Evaluation

The experimental objective is to validate that execution should occur only under admissible transitions with real-world confirmation. Baseline execution occurs after structural alignment, while proposed execution requires candidate validation, survival, reality confirmation, and collapse guard clearance.

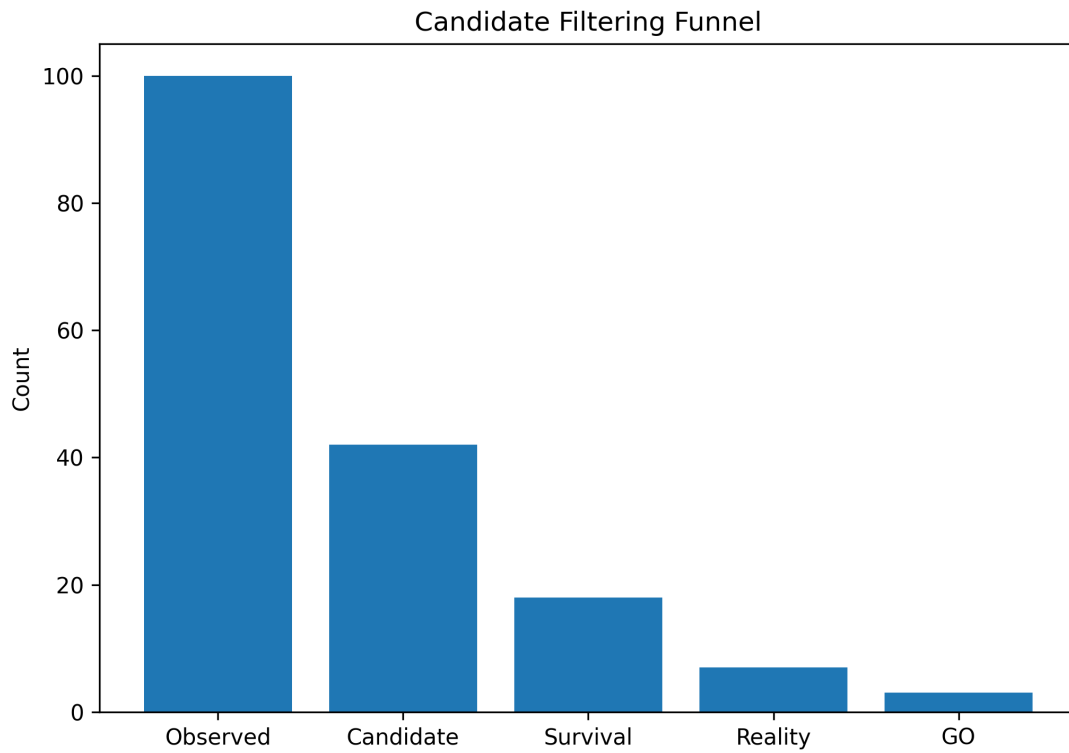


Figure 5. Candidate Filtering Funnel.

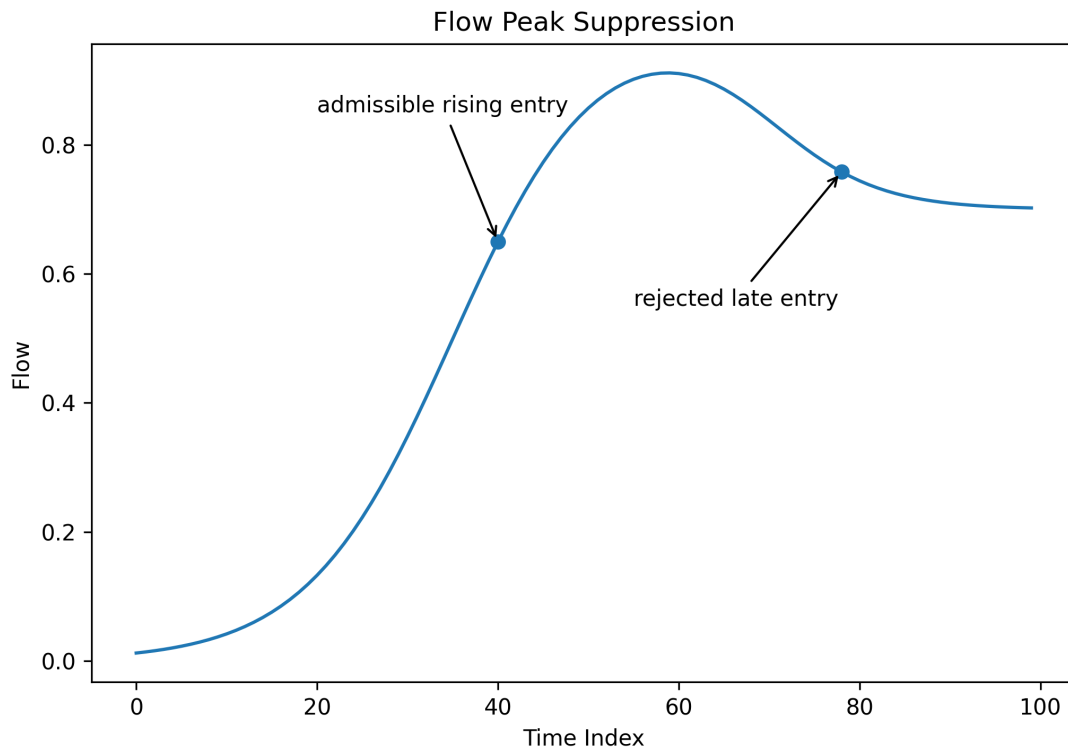


Figure 6. Flow Peak Suppression.

6. Use Case: LLM Hallucination Suppression

Hallucination is reformulated as an unsupported state transition from context to output. The LLM output is admissible only if structural consistency, grounding, causal continuity, scope boundedness, and uncertainty preservation hold.

LLM Structural Admissibility Pipeline

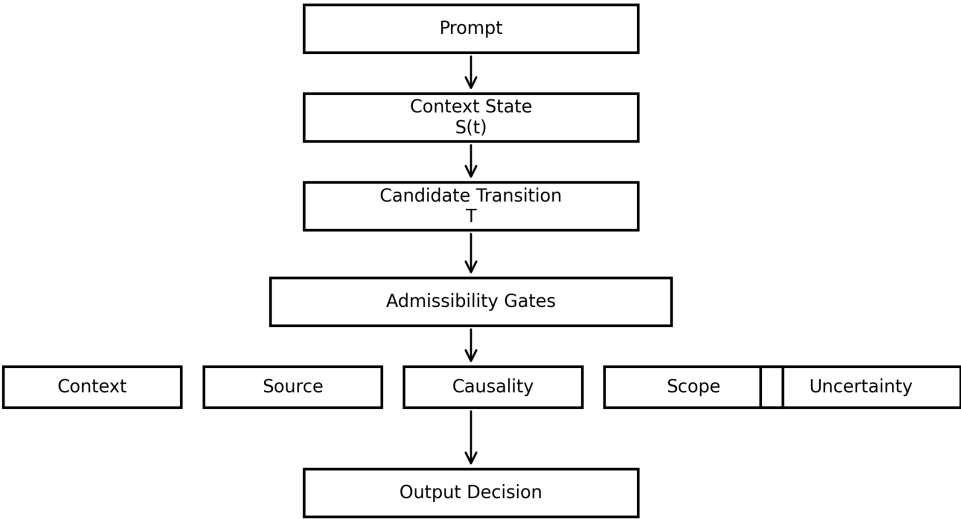


Figure 9. LLM Structural Admissibility Pipeline.

7. Conclusion

Structural Admissibility establishes a proactive security model: Security = Admissible Transition Only.